Guillaume Le Gall

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RESEARCH INTERESTS

Ph.D. in urban solar radiation variability modelling and analysis from Université Savoie Mont Blanc. Graduate engineer in optics (Institut d'Optique, Université Paris-Saclay) with an M.Sc. in light and lighting applied to the built environment (Bartlett School, University College London). My research interests span across urban solar energy, radiative transfer, Monte Carlo modelling and (multi)linear modal decomposition.

EDUCATION

2022 - 2025 **UNIVERSITÉ SAVOIE MONT BLANC | Ph.D. in Energetics and Process Engineering**
Laboratoire Procédés Énergie Bâtiment, UMR CNRS 5271, Le Bourget-du-Lac, France

Thesis. *Characterisation and modelling of solar resource variability in urban environments* [Non-amended version]. Advised by Julien Ramousse and Martin Thebault.

2020 - 2021 **UNIVERSITY COLLEGE LONDON | M.Sc. in Light and Lighting, with Distinction**
The Bartlett School of Environment, Energy and Resources, London, United Kingdom

Thesis. *Parametric design to explore the potential of curved-line folding for optimising the daylight performances of adaptative building façades.* Advised by Farhang Tahmasebi and Kevin Mansfield.
GPA. 3.95/4.00 cumulative (2/23 students)

2018 - 2021 **INSTITUT D'OPTIQUE | Dipl.-Ing. in Optics and Photonics**
Université Paris-Saclay, Palaiseau, France

GPA. 3.80/4.00 cumulative (top third students)

RESEARCH EXPERIENCE

2024 **VISITING PH.D. STUDENT | University of Bologna**
Department of Mathematics, Bologna, Italy

Three months visit to work on low-rank approximation and higher-order numerical analysis methods. Advised by Valeria Simoncini.

2022 **LIGHTING CONSULTANT | Building Research Establishment**
Watford, United Kingdom

Involvement in consultancy and research projects in the broad area of (day)lighting for private clients and public institutions.

2020 **RESEARCH INTERN | École Nationale des Travaux Publics de l'État**
Laboratoire Génie Civil et Bâtiment, UMR CNRS 5513, Vaulx-en-Velin, France

Four-months postgraduate internship (M1) within the scope of the PERCILUM project (ANR grant 19-CE38-0010) on hyperspectral data acquisition, processing and analysis for improving the visual perception of digitally-visualised lit urban scenes.

TEACHING EXPERIENCE

2022 - 2025 **PART-TIME LECTURER | Polytech Annecy-Chambéry**
Université Savoie Mont Blanc, Le Bourget-du-Lac, France

GECH615-780 *Physique du bâtiment - Éclairage pour l'environnement bâti* (Undegrad. 3/Postgrad. 1)

2022 - 2025 **TEACHING ASSISTANT | Solar Academy Graduate School**
Université Savoie Mont Blanc, Le Bourget-du-Lac, France

ENER751 *Radiative transfer* (Postgrad. 1)

ENER858 *Advanced radiation modelling in complex media* (Postgrad. 1)

2021 - 2023 **PRIVATE TEACHER | The French Tutors**
Lycée Français Charles-de-Gaulle, London, United Kingdom

Mathematics and physics (High school)

RESEARCH PUBLICATIONS

JOURNAL ARTICLES

Le Gall, G., Thebault, M., Simoncini, V., & Ramousse, J. 2025. Detailed spatiotemporal analysis of shortwave radiation variability in an urban context using higher-order orthogonal decomposition [Under review]. *Int. J. Therm. Sci.*

Le Gall, G., Thebault, M., Caliot, C., & Ramousse, J. 2025. Modal decomposition for the analysis of solar radiation variability in urban areas. *Int. J. Heat Mass Transf.*, **251**, 127316.

CONFERENCE PAPERS

Le Gall, G., Thebault, M., & Ramousse, J. 2025. Higher-order decomposition of the urban solar resource variability in space and time. *J. Phys.: Conf. Ser.*, **3140**, 032004.

Le Gall, G., Thebault, M., Simoncini, V., & Ramousse, J. 2025. Approche modale multilinéaire pour la décomposition spatio-temporelle du champ de rayonnement en milieu urbain. *In: Actes du 33ème Congrès Français de Thermique, Chambéry, France, 03–06 Juin*, 463-470.

Le Gall, G., Thebault, M., Caliot, C., & Ramousse, J. 2023. Principal Component Analysis for the characterisation of spatiotemporal variations of the solar resource in urban environments. *In: Proceedings of the 17th International Heat Transfer Conference, Cape Town, South Africa, 14–18 August.* Begell House.

OTHER PUBLICATIONS

Ticleanu, C., Howlett, G., Flores Villa, L., & **Le Gall, G.** 2023. *Daylight calculation methods in BS EN 17037.* CIBSE/BRE Research Insight 07. Chartered Institution of Building Services Engineers, London, United Kingdom.

SELECTED SEMINARS AND ORAL CONFERENCE PRESENTATIONS

2025 **CISBAT 2025**, EPFL, Lausanne, Switzerland
Higher-order modal decomposition of the solar resource variability in space and time.

11th International Symposium on Radiative Transfer (RAD-25), Kuşadasi, Turkey
Higher-order modal decomposition of urban irradiance variations in space and time.

Solar Academy Conference Series, Le Bourget-du-Lac, France
Characterisation and modelling of shortwave radiation variability on the urban scale.

2024 **Journées Nationales de l'Énergie Solaire (JNES 2024)**, Anglet, France
La décomposition modale multi-dimensionnelle comme outil d'analyse de la variabilité de la ressource solaire en milieu urbain.

Seminari del Dipartimento di Matematica, University of Bologna, Bologna, Italy
Spatiotemporal variability of solar radiation within an urban context: A characterisation by means of Principal Component Analysis.

Hélios 3rd Consortium Meeting, NTNU, Trondheim, Norway
Detailed investigation of the urban solar resource variability using modal decomposition.

COMPUTATIONAL SKILLS AND LANGUAGE PROFICIENCY

Programming and Data Processing
Python, MATLAB (incl. Deep Learning Toolbox), C/C++ (basic knowledge)

Radiation Modelling
Radiance. Significant experience acquired during my M.Sc. (UCL), Ph.D. and teaching experience (USMB) for the rendering and physics-based analysis of solar radiation propagation in urban scenes.

`htrdr-urban`. Extensive use during my Ph.D. at USMB for the detailed modelling of urban radiative transfer in the presence of complex geometries and vegetation.

Miscellaneous Software and Infrastructure
Rhino3D (including Grasshopper), UNIX environment, git, LaTex, Inkscape

Languages
French (native), English (C1 certified - TOEIC 990/990), Japanese (A1 certified), German (basic knowledge)